

How to give read and write permissions using Unix commands. A very simple example is to create a text file with a C program, then use `chmod`.

Create a file called `permission.c`

```
#include <stdio.h>
int
main()
{
    FILE *fp;    fp =
    fopen("sample.txt", "w");    if
    (fp == NULL)
    {        printf("Cannot open
    file\n");        return 1;
    }    fprintf(fp, "Hello, this is a test
    file.\n");    fclose(fp);    printf("File
    created successfully.\n");    return 0;
}
```

Compile `gcc permission.c -o permission`

Run

`./permission`

Output

File created successfully.

A file called sample.txt is created.

Check the current permissions

```
ls -l sample.txt
```

You may see

```
-rw-r--r-- 1 user user 31 Aug 17 08:20 sample.txt
```

The owner has: Read + Write

Group has: Read Others

have: Read

This is permission 644.

Give Read and Write permission to everyone `chmod`

```
666 sample.txt
```

Check `ls -l sample.txt`

You should see

```
-rw-rw-rw- 1 user user ... sample.txt
```

Instead of numbers, you can directly give:

```
chmod u+rw sample.txt chmod  
o+rw sample.txt chmod a+rw  
sample.txt
```

Task of the Day

- 1) Give Read and Write only to the owner
- 2) Give Read and Write to the owner and group
- 3) Only owner can read and write
- 4) Owner read/write, everyone else read-only
- 5) Owner can do everything, others can only read and execute
- 6) Owner and group can read/write
- 7) Only owner can execute

Fill in the permission numbers for the following:

- 8) Owner: read + write; Group: nothing; Others: nothing
→ 600
- 9) Owner: read + write; Group: read; Others: read
→ 644
- 10) Owner: read + write + execute; Group: read + execute;
Others: read + execute → 755
- 11) Owner: read + write; Group: read + write; Others:
nothing → 660
- 12) Everyone: read only → 444

```
hari@LAPTOP-C6JDL02:~$ gedit permission.c

(gedit:821): Glib-WARNING **: 05:46:47.025: Failed to read portal settings: GDBus.Error:org.freedesktop.DBus.Error.ServiceUnknown: The name org.freedesktop.p
ortal.Desktop was not provided by any .service files

(gedit:821): IBUS-WARNING **: 05:46:46.802: Failed to mkdir /home/hari/snap/gedit/697/.config/ibus/bus: Not a directory

(gedit:821): Glib-WARNING **: 05:47:13.050: Tried to map a popup with a non-top most parent
hari@LAPTOP-C6JDL02:~$ gcc permission.c -o permission
hari@LAPTOP-C6JDL02:~$ ./permission
File created successfully.
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rw-r--r-- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod 666 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rw-rw-rw- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod u+rx
chmod: missing operand after 'u+rx'
Try 'chmod --help' for more information.
hari@LAPTOP-C6JDL02:~$ chmod u+rx sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rwxrwxrwx- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ task 1
hari@LAPTOP-C6JDL02:~$ chmod u+rw sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rwxrwxrwx- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod 666 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rw-rw-rw- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ task 1
hari@LAPTOP-C6JDL02:~$ chmod 600 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rw----- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ task 2
hari@LAPTOP-C6JDL02:~$ chmod 666 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rw-rw-rw- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod 600 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rw----- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod 644
chmod: missing operand after '644'
Try 'chmod --help' for more information.
hari@LAPTOP-C6JDL02:~$ chmod 644 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rw-r--r-- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod 755 sample.txt
```

```
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rwxrwxrwx- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod 666 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rw-rw-rw- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ task 1
hari@LAPTOP-C6JDL02:~$ chmod 600 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rw----- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ task 2
hari@LAPTOP-C6JDL02:~$ chmod 666 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rw-rw-rw- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod 600 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rw----- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod 644
chmod: missing operand after '644'
Try 'chmod --help' for more information.
hari@LAPTOP-C6JDL02:~$ chmod 644 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rw-r--r-- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod 755 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rwxrwxrwx- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod 66 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-----rw- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod 755 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rwxrwxrwx- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod u+rw sample.txt
hari@LAPTOP-C6JDL02:~$ chmod g+rw sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rwxrwxrwx- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$ chmod 766 sample.txt
hari@LAPTOP-C6JDL02:~$ ls -l sample.txt
-rwxrwxrwx- 1 hari hari 29 Aug 17 05:01 sample.txt
hari@LAPTOP-C6JDL02:~$
```